

Shallow Pump

The shallow buoyancy pump is a **single-stroke piston** design that moves **seawater directly** into and out of a short port on the nose centerline (the stagnation point) to change the vehicle's displacement. It uses a **90-watt motor** and a **rolling-diaphragm (Bellofram) seal**. On G3 gliders the pump is high-displacement ($\geq \sim 960$ cc).

Source

Paraphrased and consolidated from the *Slocum G3 Glider Operators Manual*, the TWR user forum, and the UG2 community Slack. Pumps are **factory configured** — contact glidersupport@teledyne.com before changing pump settings. See also the [Deep Pump](#).

Depth / gearbox variants

The shallow pump ships with different **gearbox/motor** combinations rated for different pressures (commonly **30 / 50 / 100 / 150 / 200 m**). The mechanical gear drive is not the limiting factor — the limit is the energy that must be pulled from the battery on the return stroke. Pick the gearbox/motor for your working depth to get **quick inflections** (more important in shallow water) and to minimise return-stroke energy.

100 m vs 200 m — which to buy

Both are piston pumps. Choose the **200 m** pump if the water is deeper than 100 m and you want to sample below 100 m. The only advantage of the **100 m** pump is slightly more efficient flight (slightly longer missions); choose it only if limiting profiles to 100 m is acceptable *and* maximum endurance is the priority.